

Project Title: Flappy Goose

Overview: This project is inspired by Flappy Bird. The player controls a goose flying through KAIST-themed monument obstacles. The goose is affected by gravity, and the player presses a space to flap upward. The game is divided into stages. The first stage uses normal gravity. After each stage, the game enters a short transition phase where the goose is temporarily immortal, no obstacles appear, and the next stage receives a randomly selected gravity value. The player must adapt to the new gravity and survive as long as possible.

Requirements:

1. The game will open a graphical game window.
2. The game will begin at a home screen before gameplay starts.
3. The home screen will display the game title Flappy Goose.
4. The home screen will display the current high score.
5. The home screen will display an instruction telling the player which key starts the game.
6. When the player presses the start key from the home screen, a new game will begin.
7. Starting a new game from the home screen will reset the current score, stage number, gravity, goose position, and obstacles to their initial gameplay values.
8. The game window will display a goose character, KAIST-themed monument obstacles, a background, the current score, and the current stage number during gameplay.
9. The player will control the goose by pressing the Space key.
10. When the player presses Space, the goose will receive an upward velocity.
11. When the player does not press Space, gravity will pull the goose downward.
12. The goose's vertical position will continuously update during gameplay according to its current velocity and the current gravity value.
13. The first stage will use a normal gravity value.
14. Each stage will contain a fixed number of monument obstacles that move from right to left.
15. Each monument obstacle will contain one open gap that the goose can pass through.

16. Each obstacle will be visually represented as a KAIST monument-inspired image.
17. Collision detection for each obstacle will use two invisible rectangles: one rectangle above the gap and one rectangle below the gap.
18. The goose will gain one point when it successfully passes an obstacle.
19. The player loses if the goose collides with an obstacle during normal gameplay.
20. The player loses if the goose moves outside the top or bottom boundary of the game window during normal gameplay.
21. After the goose passes all obstacles in a stage, the game will enter a transition phase.
22. During the transition phase, no new obstacles will appear.
23. During the transition phase, the goose will be immortal and cannot lose by colliding with obstacles or screen boundaries.
24. The transition phase will last approximately 3 seconds.
25. At the start of each transition phase, the game will randomly select the gravity value for the next stage from a predefined safe range.
26. During the transition phase, the game will display a message indicating that a gravity shift is happening.
27. During the transition phase, the game will display a countdown before the next stage begins.
28. After the transition phase ends, the next stage will begin using the newly selected gravity value.
29. The obstacle gap size will be large enough for the goose to pass through.
30. The obstacle gap position will be generated within safe vertical bounds so that the gap is not too close to the top or bottom edge of the window.
31. Consecutive obstacle gaps will not move too far vertically from one obstacle to the next, so the level remains reasonably passable.
32. The horizontal speed of obstacles may increase gradually as the stage number increases.
33. When the player loses, the game will compare the player's final score with the current high score.
34. If the final score is greater than the current high score, the game will update the high score to the final score.
35. If the final score is greater than the previous high score, the game will display a special new-high-score game-over screen.

36. The special new-high-score game-over screen will congratulate the player and display both the final score and the new high score.
37. If the final score is not greater than the current high score, the game will display a normal game-over screen.
38. The normal game-over screen will display the final score and the current high score.
39. From either game-over screen, the player will be able to return to the home screen by pressing a key.
40. When returning to the home screen after game over, the home screen will display the updated high score.
41. From either game-over screen, the player will be able to restart the game by pressing a key.
42. Restarting the game after game over will reset the current score, stage number, gravity, goose position, and obstacles to the initial gameplay state, but it will not reset the high score.
43. The high score only needs to be maintained while the program is running.
44. Closing and reopening the program may reset the high score to zero.
45. The game will be runnable from the public GitHub repository using the instructions in README.md.

Example Interaction: The game starts at a home screen showing the title Flappy Goose, the current high score, and an instruction to start the game. The player presses the start key, and the first stage begins. A goose appears on the left side of the screen, and KAIST-themed monument obstacles move from right to left. The player presses Space to make the goose flap upward. Gravity pulls the goose downward between flaps. The player guides the goose through the gap in each monument obstacle and gains one point for each obstacle passed.

After the goose passes all obstacles in the first stage, the game enters a transition phase. The screen displays a message such as “Gravity Shift!” and a countdown. During this transition, no obstacles appear and the goose cannot die. After approximately 3 seconds, the next stage begins with a newly selected random gravity value. The player continues passing obstacles until the goose collides with an obstacle or leaves the screen during normal gameplay.

When the game ends, the system compares the final score with the current high score. If the final score is higher than the previous high score, the game updates the high score and displays a special congratulation screen showing the new high score. If the final score is not higher than the current high score, the game displays a normal game-

over screen showing the final score and the high score. The player can then restart the game or return to the home screen, where the current high score is displayed.